

TECHIN

JavaScript

Strings

JavaScript Strings
(darbas su tekstais)

Kėlimas į naują eilutę, kai naudojame paprastas kabutes.

```
1  let guestList = "Guests:\n * John\n * Pete\n * Mary";
2
3  alert(guestList); // a multiline list of guests
```

Specialieji simboliai:

Character	Description
`\n`	New line
`\r`	In Windows text files a combination of two characters `\r\n` represents a new break, while on non-Windows OS it's just `\n` . That's for historical reasons, most Windows software also understands `\n` .
`\'`, `\"`	Quotes
`\\`	Backslash
`\t`	Tab
`\b`, `\f`, `\v`	Backspace, Form Feed, Vertical Tab – kept for compatibility, not used nowadays.



JavaScript Strings
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Kabučių pažymėjimas "escape", kai naudojam paprastas kabutes

```
alert( 'I\'m the Walrus!' ); // I'm the Walrus!
```

Escape simboliai:

Character	Description
`\n`	New line
`\r`	In Windows text files a combination of two characters `\r\n` represents a new break, while on non-Windows OS it's just `\n`. That's for historical reasons, most Windows software also understands `\n`.
`\'`, `\"`	Quotes
`\\`	Backslash
`\t`	Tab
`\b`, `\f`, `\v`	Backspace, Form Feed, Vertical Tab – kept for compatibility, not used nowadays.



JavaScript Strings
(darbas su tekstais)

Kėlimas į naują eilutę, kai naudojame backticks.

```
// two lines using a normal newline and backticks
let str2 = `Hello
World`;
```

Naudojant backticks kabutės escape simboliu nežymimos:

```
alert( `I'm the Walrus!` ); // I'm the Walrus!
```

Kabutės žymimos tik tada, kai string'ui aprašyti naudojamos kabutės sutampa su tekste esančiomis kabutėmis.



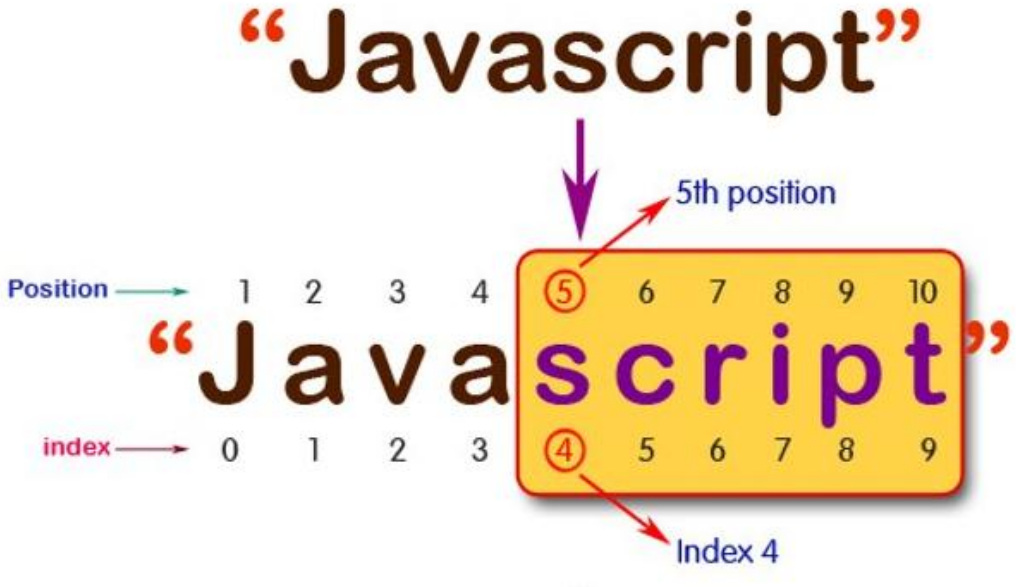
JavaScript Strings
(darbas su tekstais)

Teksto ilgio simboliais radimas:

```
<script>
"use strict";
let str ="My name"
alert( str.length ); // 7
</script>
```

JavaScript Strings
(darbas su tekstais)

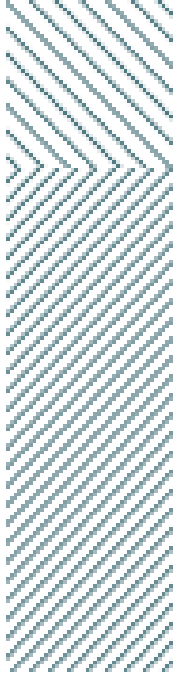
Kiekvienas teksto simbolis ir tarpas turi index'ą.



JavaScript Strings
(darbas su tekstais)

Teksto simbolių "pasiekimas"

```
1  let str = `Hello`;
2
3  // the first character
4  alert( str[0] ); // H
5  alert( str.charAt(0) ); // H
6
7  // the last character
8  alert( str[str.length - 1] ); // o
```





JavaScript Strings
(darbas su tekstais)

String yra nekeičiami (immutable).
Tiesiogiai keisti tekstų negalime.

```
1  let str = 'Hi';
2
3  str[0] = 'h'; // error
4  alert( str[0] ); // doesn't work
```



Reikia kurti naujus, naudoti sujungimą.

```
1  let str = 'Hi';
2
3  str = 'h' + str[1]; // replace the string
4
5  alert( str ); // hi
```





JavaScript Strings

Dažniausiai naudojami String metodai

Method	Description
charAt(index)	returns the character at the specified index
concat()	joins two or more strings
replace()	replaces a string with another string
split()	converts the string to an array of strings
substr(start, length)	returns a part of a string
substring(start,end)	returns a part of a string
slice(start, end)	returns a part of a string
toLowerCase()	returns the passed string in lower case
toUpperCase()	returns the passed string in upper case
trim()	removes whitespace from the strings
includes()	searches for a string and returns a boolean value
search()	searches for a string and returns a position of a match



JavaScript Strings

charAt() metodas grąžina simbolį iš nurodytos pozicijos index. Grąžinama reikšmė yra string tipo.

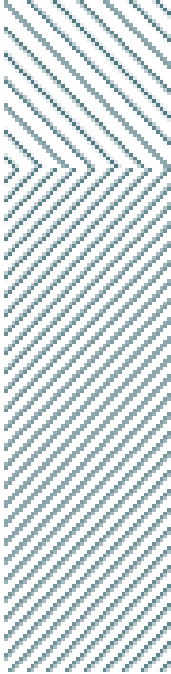
```
let text = "HELLO WORLD";
let letter = text.charAt(1);
```

```
let text = "HELLO WORLD";
let letter = text.charAt(text.length-1);
```

JavaScript Strings

concat() sujungia du tekstus.

```
let text1 = "Hello";
let text2 = "world!";
let result = text1.concat(" ", text2);
```





JavaScript Strings

replace() skirtas tekstams pakeisti

```
let text = "Visit Microsoft!";
let result = text.replace("Microsoft", "W3Schools");
```

Pavyzdyje pateiktas skriptas suras tekste Microsoft ir pakeis į W3Schools.
Jei žodžių yra keli, pakeis pirmąjį rastą.

Jei norime pakeisti visus atitikmenis, naudojame replaceAll()

```
let result = "1 abc 2 abc 3".replaceAll("abc", "xyz");
// `result` is "1 xyz 2 xyz 3"
```



JavaScript Strings

split() skirtas tekstams išskaidyti į masyvą

```
const greeting = "What a beautiful world";
const arr = greeting.split(' ');
console.log(arr);
```

```
["What", "a", "beautiful", "world"]
```

splitter

```
const greeting = "What a beautiful world";
const arr = greeting.split(''); // The splitter is an empty string,
console.log(arr);
```

```
['W', 'h', 'a', 't', ' ', 'a', ' ', 'b', 'e', 'a', 'u', 't', 'i', 'f
```

JavaScript Strings

split() metodas dažnai naudojamas kartu su masyvo metodu join()

```
const sentence = "Hello world this is JavaScript";

// Split the sentence into an array of words
const words = sentence.split(" ");
console.log(words);
// Output: ["Hello", "world", "this", "is", "JavaScript"]

// Join the array back into a sentence with dashes
const dashed = words.join("-");
console.log(dashed);
// Output: "Hello-world-this-is-JavaScript"
```



JavaScript Strings

substring() ir slice() grąžina nurodytą teksto dalį, esantį tarp nurodytų indeksų.

```
str.substring(indexStart, indexEnd)
```

```
const message = "JavaScript is fun.";

// get the substring starting from index 0 to 10
let result = message.substring(0, 10);
console.log(result);

// Output: JavaScript
```

```
str.slice(beginIndex, endIndex)
```

```
const str = "JavaScript is a very absurd programming language.";

// from index 28 to end
console.log(str.slice(28)); // 'programming language.'

// from index 4 to 14
console.log(str.slice(4, 15)); // 'Script is a'
```

Skirtumai: https://thispointer.com/difference-between-slice-and-substring-in-javascript/



JavaScript Strings

includes() patikrina, ar nurodytas tekstas yra kito teksto dalis, grąžina true arba false

```
let sentence = "Java is to JavaScript what Car is to Carpet.";

let check = sentence.includes("Java");
console.log(check); // true

// case sensitive
let check1 = sentence.includes("java");
console.log(check1); // false

// second argument specifies position to start at
let check2 = sentence.includes("Java", 20);
console.log(check2); // false

let check3 = sentence.includes("whose");
console.log(check3); // false

let check4 = sentence.includes("");
console.log(check4); // true
```



JavaScript Strings

search() ieško regex atitikties tekste.
Grąžina pirmosios atitikties index, arba -1, jei atitikimų regex'ui nerado.

```
// defining string
let string1 = "JavaScript JavaScript1";

// pattern having 'JavaScript' followed by a digit
let regExp = /(JavaScript)\d/;

// searching for a match between regExp and given string
let index = string1.search(regExp);

console.log(index);
```

Daugiau string metodų:

https://www.programiz.com/javascript/library/string

https://www.javascripttutorial.net/javascript-string-methods/

https://youtu.be/x16Z_U1tGrc?si=G66nv7p857pqIalk

